

Multimodal RAG Knowledge Guide

A multimodal retrieval-augmented generation system indexes evidence from prose, tables, charts, and images. Each modality is converted into searchable documents while retaining page and content-type metadata.

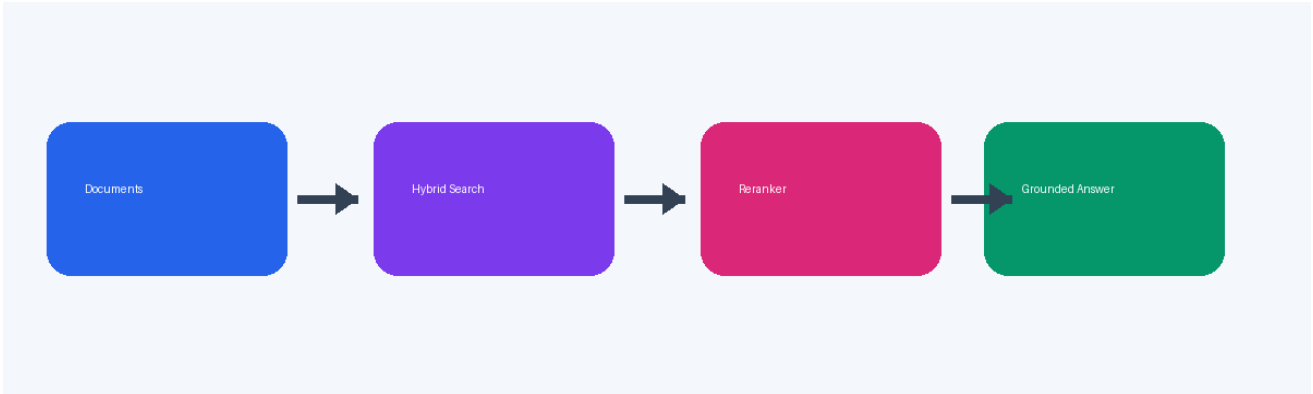


Figure 1. A two-stage RAG pipeline uses hybrid candidate retrieval followed by semantic reranking and grounded answer generation.

Retrieval Comparison

Method	Semantic Match	Exact Terms	Typical Role
Dense vectors	High	Medium	Candidate retrieval
BM25 sparse	Medium	High	Keyword retrieval
Hybrid fusion	High	High	Candidate combination
Cross-encoder	Very high	High	Final reranking

Table 1. Dense and sparse retrieval contribute complementary signals; a cross-encoder provides the most detailed pairwise relevance judgment.

Illustrative Quality Chart

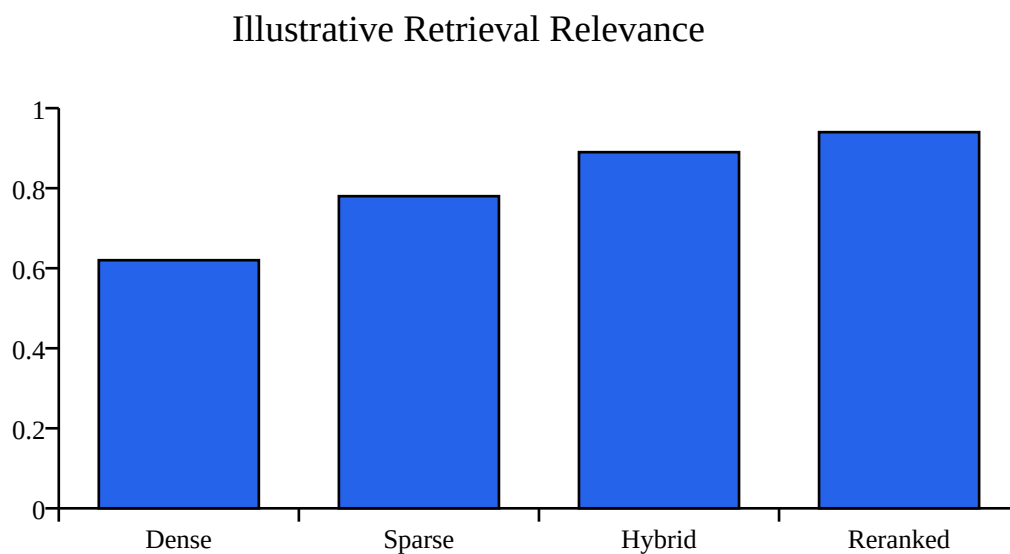


Figure 2. Illustrative relevance scores rise when dense and sparse retrieval are fused and then reranked. Values demonstrate the concept and are not benchmark results.

How Hybrid Search Works

Dense retrieval finds passages with similar meaning even when wording differs. BM25 rewards exact and rare query terms. Reciprocal Rank Fusion combines their ranked lists without assuming their scores share a scale. A cross-encoder then evaluates the question and each fused candidate jointly.

Multimodal Indexing

Text is stored directly. Tables are serialized with column headers so relationships between cells remain visible. Images and charts are indexed with captions, page numbers, dimensions, and nearby explanatory text. This normalization allows a text model to retrieve evidence originating from visual content.